MsrA, LG+C60+F+G (PMSF)
groups of one taxon collapsed; nodes with ultrafast bootstrap < 50 collapsed; supports >= 70 shown

length (aa)
Chlamydomonas_reinhardtii_gi_23452038_gb 160
Blan_MsrA.2 (selenocysteine) 160
Bflo_MsrA.2 (selenocysteine) 168
Fbor_MsrA.2.cysteine 155
Fbor_MsrA.8.cysteine 132
Fbor_MsrA.6.cysteine 155
Bsty_MsrA.1.cysteine 152
Mery_MsrA.1.cysteine 165
Bsty_MsrA.2.cysteine 169
Fbor_MsrA.17 (unaligned) 144
Fbor_MsrA.16.cysteine 165
Fbor_MsrA.19 (unaligned) 81
Fbor_MsrA.3.cysteine 182
Fbor_MsrA.7.cysteine 183
Fbor_MsrA.24 (unaligned) 110
Fbor_MsrA.23 (unaligned) 126
Fbor_MsrA.20.cysteine 87
Fbor_MsrA.25.cysteine 66
Fbor_MsrA.33.cysteine 134
Fbor_MsrA.26.cysteine 106
Fbor_MsrA.22.cysteine 76
Fbor_MsrA.11.cysteine 178
Fbor_MsrA.12.cysteine 183
Fbor_MsrA.27.cysteine 94
Fbor_MsrA.18.cysteine 151
Streptomyces_pristinaespiralis_ATCC_2548 220
Streptomyces_clavuligerus_ATCC_27064_gi_ 224
Anaeromyxobacter_dehalogenans_2CP-1_gi_2 213
Anaeromyxobacter_sp._Fw109-5_gi_15300420 216
Synechococcus_elongatus_PCC_6301_gi_5675 246
Endoriftia_persephone_Hot96_1_Hot96_2_gi 212
Magnetospirillum_gryphiswaldense_MSR-1_g 209
Thioalkalivibrio_sp._HL-EbGR7_gi_2209363 219
Rhodospirillum_centenum_SW_gi_209964138_ 219
Plesiocystis_pacifica_SIR-1_gi_149917293 222
gamma_proteobacterium_NOR5-3_gi_21967742 218
gamma_proteobacterium_KT_71_gi_88705939_ 218
Frankia_sp._EAN1pec_gi_158312691_ref_YP_ 221
Frankia_alni_ACN14a_gi_111225545_ref_YP_ 233
Methylococcus_capsulatus_str._Bath_gi_53 222
Acidiphilium_cryptum_JF-5_gi_148259423_r 225
Saccharopolyspora_erythraea_NRRL_2338_gi 227
Rhodococcus_jostii_RHA1_gi_111020994_ref 220
Mycobacterium_abscessus_gi_169627225_ref 248
marine_gamma_proteobacterium_HTCC2207_gi 207
Stenotrophomonas_maltophilia_K279a_gi_19 228
Xanthomonas_oryzae_pv._oryzae_MAFF_31101 216
Xylella_fastidiosa_Ann-1_gi_71898515_ref 216
Acidothermus_cellulolyticus_11B_gi_11792 212
Thermobifida_fusca_YX_gi_72163077_ref_YP 222
Salinispora_tropica_CNB-440_gi_145593476 220
Kineococcus_radiotolerans_SRS30216_gi_15 223
Fbor_MsrA.1.cysteine 142
Maricaulis_maris_MCS10_gi_114568707_ref_ 217
Chromohalobacter_salexigens_DSM_3043_gi_ 211
Marinomonas_sp._MWYL1_gi_152995879_ref_Y 219
Labrenzia_alexandrii_DFL-11_gi_224397435 216
Stappia_aggregata_IAM_12614_gi_118589621 221
gi_20455117_sp_Q93S39.1_MSRA_OCHAN_gi_20 218
Agrobacterium_tumefaciens_str._C58_gi_15 216
Mesorhizobium_sp._BNC1_gi_110636305_ref_ 219
Beijerinckia_indica_subsp._indica_ATCC_9 218
Hyphomonas_neptunium_ATCC_15444_gi_11479 216
Azorhizobium_caulinodans_ORS_571_gi_1584 218
Parvibaculum_lavamentivorans_DS-1_gi_154 218
Magnetospirillum_magnetotacticum_MS-1_gi 219
Rhodopseudomonas_palustris_CGA009_gi_399 218
Bradyrhizobium_sp._BTAi1_gi_148259018_re 218
Rhodopseudomonas_palustris_BisA53_gi_115 218
Phenylobacterium_zucineum_HLK1_gi_197106 217
Brevundimonas_sp._BAL3_gi_223484811_ref_ 216
Caulobacter_crescentus_CB15_gi_16125246_ 217
Rhodobacter_sphaeroides_ATCC_17029_gi_12 224
Rhodobacterales_bacterium_HTCC2654_gi_84 215
Loktanella_vestfoldensis_SKA53_gi_845166 216
Dinoroseobacter_shibae_DFL_12_gi_1590436 220
Pseudomonas_syringae_pv._syringae_B728a_ 215
Pseudomonas_entomophila_L48_gi_104784061 222
Pseudomonas_mendocina_ymp_gi_146305609_r 219
Pseudomonas_aeruginosa_PA7_gi_152986954_ 215
Azotobacter_vinelandii_AvOP_gi_67158063_ 216
Pseudomonas_stutzeri_A1501_gi_146284168_ 245
Proteus_mirabilis_HI4320_gi_197287201_re 212
Salmonella_enterica_subsp._enterica_sero 212
Serratia_proteamaculans_568_gi_157368705 213
Photobacterium_profundum_3TCK_gi_9041196 211
Vibrio_vulnificus_YJ016_gi_37678611_ref_ 209
Vibrio_cholerae_MZO-3_gi_153801142_ref_Z 212
Vibrio_parahaemolyticus_16_gi_219549712_ 209
Vibrionales_bacterium_SWAT-3_gi_14897779 209
Vibrio_campbellii_AND4_gi_163802638_ref_ 212
Vibrio_sp._Ex25_gi_194538879_ref_YP_0020 212
Arthrospira_maxima_CS-328_gi_209525158_r 219
Synechococcus_sp._PCC_7335_gi_196192362_ 221
Synechocystis_sp._PCC_6803_gi_16329216_r 222
marine_gamma_proteobacterium_HTCC2143_gi 215
Gloeobacter_violaceus_PCC_7421_gi_375220 221
Fbor_MsrA.10.cysteine 82
Saccharophagus_degradans_2-40_gi_9002150 215
marine_gamma_proteobacterium_HTCC2148_gi 226
Fulvimarina_pelagi_HTCC2506_gi_114706079 222
Aurantimonas_sp._SI85-9A1_gi_90419745_re 223
marine_gamma_proteobacterium_HTCC2080_gi 243
uncultured_marine_bacterium_311_gi_40062 213
alpha_proteobacterium_HTCC2255_gi_114771 215
gamma_proteobacterium_NOR51-B_gi_2196802 221
Nodularia_spumigena_CCY9414_gi_119510530 222
Acaryochloris_marina_MBIC11017_gi_158335 218
Microcystis_aeruginosa_NIES-843_gi_16636 220
Cyanothece_sp._PCC_7424_gi_218441887_ref 220
Cyanothece_sp._PCC_7425_gi_220907947_ref 220
Nitrosococcus_oceani_ATCC_19707_gi_77165 219
Microcoleus_chthonoplastes_PCC_7420_gi_2 220
Lyngbya_sp._PCC_8106_gi_119485559_ref_ZP 222
Synechococcus_sp._JA-2-3B_a_2-13_gi_8660 222
Thermosynechococcus_elongatus_BP-1_gi_22 224
Hror_MsrA.4.cysteine 236
Scla_MsrA.5.cysteine 173
Bsch_MsrA.2 (unaligned) 179
Stho_MsrA.1.cysteine 199
Cint_MsrA.5.cysteine 216
Pmam_MsrA.4.cysteine 242
Tdem_MsrA.5.cysteine 198
Pege_MsrA.4.cysteine 168
Xenopus_tropicalis_gi_170284912_gb_AAI60 257
Drer_MsrA.3.cysteine 235
Salmo_salar_gi_213514724_ref_NP_00113436 233
Taeniopygia_guttata_gi_224048943_ref_XP_ 213
Bos_taurus_gi_27806537_ref_NP_776539.1 233
Human_MSRA_NP_036463.1 235
Pan_troglodytes_gi_114623486_ref_XP_0011 191
Monodelphis_domestica_gi_126304193_ref_X 231
Mammals (2) 234 (233-234)
Xenopus_Silurana_tropicalis_gi_166795959 207
Oceanobacter_sp._RED65_gi_94500335_ref_Z 206
Ray-finned fishes (2) 252 (238-267)
Fbor_MsrA.4.cysteine 223
Ovan_MsrA.1 (selenocysteine) 171
Oalb_MsrA.2.cysteine 233
Oalb_MsrA.1.cysteine 196
OKI_MsrA.6.cysteine 236
OdB3_MsrA.1.cysteine 233
OSKA_MsrA.4.cysteine 256
Oalb_MsrA.5.cysteine 217
Ovan_MsrA.4.cysteine 213
OKI_MsrA.2.cysteine 217
OdB3_MsrA.3.cysteine 217
OSKA_MsrA.2.cysteine 217
Olon_MsrA.4.histidine 113
Bsty_MsrA.3 (unaligned) 109
Olon_MsrA.3.cysteine 220
Olon_MsrA.1.cysteine 300
Mery_MsrA.3.cysteine 138
0
1
2
3
4
5
6
7
8
branch length